§	Check	Result	§	Check	Result
§1	**Import & Instantiation**				**8 / 8**
1	`PortfolioBudgetingEnv` importable	*pass*	1	Env instantiation (fallback config)	*pass*
1	has `observation_space`	*pass*	1	has `action_space`	*pass*
1	has `reset`	*pass*	1	has `step`	*pass*
1	has `close`	*pass*	1	has `render`	*pass*
§2	**Space Declarations**				**7 / 7**
2	`observation_space` dtype is float32	*pass*	2	`action_space` dtype is float32	*pass*
2	`action_space` low == 0	*pass*	2	`action_space` high == 1	*pass*
2	obs shape matches `observation_space`	*pass*	2	obs within `observation_space` bounds	*pass*
2	`action_space` shape == `(n_projects,)`	*pass*			
§3	**`reset()` Contract**				**13 / 13**
3	reset returns tuple of length 2	*pass*	3	obs is ndarray	*pass*
3	info is dict	*pass*	3	info has `t_episode`	*pass*
3	info has `budget`	*pass*	3	info has `horizon`	*pass*
3	info has `n_active`	*pass*	3	info has `episode_header`	*pass*
3	info has `render_state`	*pass*	3	`t_episode` == 0 after reset	*pass*
3	`budget > 0` after reset	*pass*	3	second reset works	*pass*
3	obs shape stable across resets	*pass*			
§4	**`step()` Contract**				**10 / 10**
4	step returns 5-tuple	*pass*	4	obs2 is ndarray	*pass*
4	reward is float	*pass*	4	terminated is bool	*pass*
4	truncated is bool	*pass*	4	info2 is dict	*pass*
4	obs2 shape matches space	*pass*	4	obs2 within space	*pass*
4	info2 has `cum_reward`	*pass*	4	info2 has `render_post`	*pass*
§5	**Determinism**				**2 / 2**
5	same seed → identical first obs	*pass*	5	different seeds → different obs	*pass*
§6	**Random Policy Rollout**				**6 / 6**
6	rollout completes without exception	*pass*	6	episode ends (terminated or truncated)	*pass*
6	all obs shapes identical	*pass*	6	last obs within space	*pass*
6	episode ends (terminated or truncated)	*pass*	6	episode ends (terminated or truncated)	*pass*
§7	**Zero-Action Rollout**				**1 / 1**
7	zero-action rollout completes	*pass*			
§8	**Greedy Rollout**				**1 / 1**
8	greedy rollout completes	*pass*			
§9	**Multi-Episode Stability**				**2 / 2**
9	no exceptions across episodes	*pass*	9	obs shape consistent across episodes	*pass*
§10	**`gymnasium.utils.check_env`**				**1 / 1**
10	`check_env` passed	*pass*			
Total — all checks passed					**51 / 51**

Config: `TC-3XMAIF-R`. *Generated: 2026-10-20 17:07. Verified against the Gymnasium API standard.*